Part II Convergence of Riemannian manifolds

Main reference:

Petersen - Riemannian Geometry GTM 171

Gromov - Hausdorff convergence

Defn Let (X, d) be a metric space, $A, B \subset X$. Hausdorff distance

$$d_H(A, B) = \inf \left\{ \varepsilon \;\middle|\; \begin{array}{l} A \subset B(B, \varepsilon) \\ B \subset B(A, \varepsilon) \end{array} \right\}$$

where $B(A, \varepsilon) = \left\{ x \in X \;\middle|\; \begin{array}{l} d(x, a) < \varepsilon \\ \text{for some } a \in A \end{array} \right\}$

Theorem If (X, d) is compact, then $K(X) = \left\{ \begin{array}{c} \text{compact} \\ A \subset X \end{array} \right\}$ is a compact metric space under d_H.

Definition Let X, Y be metric spaces. Gromov - Hausdorff distance

$$d_{GH}(X, Y) = \inf \left\{ d_H(X, Y) \;\middle|\; \begin{array}{l} \text{metrics on } X \sqcup Y \\ \text{extending metrics} \\ \text{on } X, Y \end{array} \right\}$$

E.g. Let $Y = \{p\}$ one point space.

X any metric space

- $d_{GH}(X, Y) \leq \frac{D}{2}$ $\qquad D = \operatorname{diam}(X) = \sup\{d_X(x_1, x_2) \mid x_i \in X\}$

To see this, define metric on $X \amalg Y$ by $d(x, p) = D/2 \quad \forall x \in X$

need to check: $\underbrace{d(x_1, p)}_{D/2} + \underbrace{d(p, x_2)}_{D/2} \geq d(x_1, x_2)$

and $p \in B(X, D/2 + \varepsilon) \quad \forall \varepsilon > 0$

$X \subset B(p, D/2 + \varepsilon)$

$\Rightarrow d_H(X, p) \leq D/2$

$\Rightarrow d_{GH}(X, Y) \leq D/2$

- $d_{GH}(X, Y) \geq D/2$, since if

d is dist. on $X \amalg Y$

and $d(x_1, x_2) > D - \varepsilon$,

and $X \subset B(p, d_{GH} + \varepsilon)$

$p \in B(X, d_{GH} + \varepsilon)$

$$\Rightarrow D - \varepsilon < d(x_1, x_2) \leq d(x_1, p) + d(p, x_2) \leq 2 d_{GH} + 2\varepsilon$$

so $d_{GH} \geq \frac{D}{2} - \frac{3}{2}\varepsilon \quad \forall \varepsilon > 0$.

<u>Proposition</u> If X, Y are compact and $d_{GH}(X,Y)=0$ then X and Y are isometric.

<u>Proof</u> For any $i>0$ we can find metric d_i on $X \sqcup Y$ s.t. $d_H(X,Y) < i^{-1}$ under this metric.

$\Longrightarrow$ find maps $f_i : X \to Y$
$g_i : Y \to X$

s.t. $d_i(x, f_i(x)) < i^{-1} \quad \forall x \in X$
$d_i(y, g_i(y)) < i^{-1} \quad \forall y \in Y$

triangle ineq $\Longrightarrow$

$$d_i(f_i(x_1), f_i(x_2)) \le d_i(f_i(x_1), x_1) + d_i(x_1, x_2) + d_i(x_2, f_i(x_2))$$

$$\Longrightarrow d_Y(f_i(x_1), f_i(x_2)) \le d_X(x_1, x_2) + 2i^{-1}$$

and $d_X(g_i(y_1), g_i(y_2)) \le d_Y(y_1, y_2) + 2i^{-1}$

for all $x_1, x_2 \in X$, $y_1, y_2 \in Y$.

$$d_i(x, g_i(f_i(x))) \le d_i(x, f_i(x)) + d_i(f_i(x), g_i(f_i(x))) \le 2i^{-1}$$

and $d_i(y, f_i(g_i(y))) \le 2i^{-1}$

We can extract limit functions
$f: X \to Y$ and $g: Y \to X$:

Let $x_1, x_2, x_3, \dots$ be a countable dense set in X

The sequence $f_1(x_1), f_2(x_1), f_3(x_1), f_4(x_1), \dots$
has convergent subsequence

$$f_{1,1}(x_1),\ f_{1,2}(x_1),\ f_{1,3}(x_1), \dots$$

choose further subsequence of $\{f_{1,1}, f_{1,2}, \dots\}$
s.t. $f_{2,1}(x_2), f_{2,2}(x_2), f_{2,3}(x_2), \dots$ converges

keep choosing subsequences s.t. we get
convergent sequences in Y :

$$f_{3,1}(x_3),\ f_{3,2}(x_3),\ f_{3,3}(x_3), \dots$$
$$f_{4,1}(x_4), \dots$$

Let $F_i = f_{i,i}$. Then $F_1(x_j), F_2(x_j), \dots$
converges for all j

Define F_∞ on $\{x_1, x_2, \dots\} \subset X$
by $F_\infty(x_j) = \lim_{i \to \infty} F_i(x_j)$

$$d_Y(F_\infty(x_i), F_\infty(x_j)) \leq d_X(x_i, x_j) \quad \forall i, j$$

$\Rightarrow F_\infty$ extends to continuous $F_\infty : X \to Y$

Similarly we can get $G_\infty : Y \to X$

we have $d_X(x, G_\infty F_\infty(x)) = 0$

so $G_\infty F_\infty = Id$. Similarly $F_\infty G_\infty = Id$.

Both F, G are dist. decreasing $\Rightarrow$ they are isometries $\square$

<u>E.g.</u> If $A \subset X$ is finite and $X \subset B(A, \varepsilon)$, then A is <u>ε-dense in X</u>

Then $d_{GH}(A, X) \leq \varepsilon$

If X is compact then $\forall \varepsilon > 0$ $\exists$ ε-dense (finite) sets $A \subset X$.

If $A = \{x_1, \dots, x_k\} \subset X$
$B = \{y_1, \dots, y_k\} \subset Y$
are ε-dense and
$$|d_X(x_i, x_j) - d_Y(y_i, y_j)| \leq \varepsilon \quad \forall i, j$$
then $d_{GH}(X, Y) \leq 3\varepsilon$.

To see this it is enough to check that $d_{GH}(A, B) \leq \varepsilon$

<u>Theorem</u> Let $\mathcal{M}$ = compact metric spaces up to isometry.

Then $(\mathcal{M}, d_{GH})$ is a complete metric space.

<u>Proof</u> (sketch)

Let $\{X_n\} \subset \mathcal{M}$ be a Cauchy sequence,

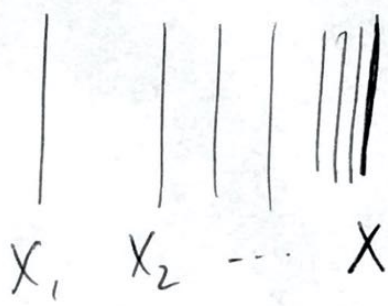
assume $d_{GH}(X_i, X_{i+1}) < 2^{-i} \quad \forall i.$

choose metrics on $X_i \sqcup X_{i+1}$ s.t. X_i, X_{i+1} are 2^{-i} Hausdorff close.

define metric on $X_i \sqcup X_{i+j}$ for $j > 1$ by

$$\underset{\substack{\uparrow \\ X_i}}{d(x_i,} \underset{\substack{\uparrow \\ X_{i+j}}}{x_{i+j})} = \min_{x_{i+k} \in X_{i+k}} \left\{ d(x_i, x_{i+1}) + \dots + d(x_{i+j-1}, x_{i+j}) \right\}$$

Let $\hat{X} = \left\{ \begin{array}{c} \text{sequences} \\ \{x_i\} \end{array} \;\middle|\; \begin{array}{l} x_i \in X_i \text{ s.t.} \\ d(x_i, x_j) \to 0 \text{ as } i,j \to \infty \end{array} \right\}$

Define $X = \hat{X}/\sim$ where

$\{x_i\} \sim \{y_i\}$ if $d(x_i, y_i) \longrightarrow 0$